1. Clinical Question & Core Hypothesis

Clinical Question

CEACAM5 is upregulated in bladder tumors and correlates with higher immune cell infiltration and MSI-H status, yet it is unknown whether this immune-rich state driven by MSI-H yields effective antitumor immunity or is functionally inhibited. 2) Tier 1 path used: tumors → CEACAM5 → higher immune cell infiltration → MSI-H (single coherent mechanistic chain linking tumor CEACAM5 expression, immune infiltration, and MSI-H). I built on this path because it directly connects a tumor-expressed adhesion/immune-interacting molecule (CEACAM5) with an immunogenic tumor phenotype (MSI-H) and observed immune infiltration. 3) Tier 2 evidence: none provided (no additional supporting paths were available), so no secondary mechanistic anchors were incorporated. 4) Endpoint choice: therapy_response was selected because MSI-H and high immune infiltration are clinically most relevant to predicting response to immune checkpoint blockade; the central translational question is whether CEACAM5 modulates that response. 5) Deliberate exclusions: ultrasound (imaging modality) was excluded because it is a non-causative clinical workflow node and does not inform tumor biology here; I excluded any genes or modifiers not present in the supplied path to avoid introducing unsupported molecular actors?

Working Hypothesis

In bladder tumors, MSI-H status drives neoantigen generation that recruits tumor-infiltrating lymphocytes (TILs) and is associated with elevated CEACAM5 expression. While high CEACAM5 correlates with increased TIL density in MSI-H tumors, CEACAM5 engagement exerts immunomodulatory effects that dampen T cell effector function. Consequently, among MSI-H, TIL-high bladder cancers, those with high CEACAM5 expression will exhibit a reduced therapy_response to immune checkpoint blockade compared with MSI-H, TIL-high tumors with low CEACAM5. This yields testable predictions: (1) CEACAM5 expression positively correlates with TIL density specifically in MSI-H bladder cancers; (2) disrupting CEACAM5 signaling enhances T cell–mediated cytotoxicity in vitro/ex vivo and improves responsiveness to PD-1/PD-L1 blockade in models; (3) clinical datasets show lower therapy_response rates to immunotherapy in CEACAM5-high/MSI-H bladder tumors compared with CEACAM5-low/MSI-H counterparts, despite similar TIL counts; (4) this pattern is specific to MSI-H contexts and not observed in MSS tumors. Secondary modifiers (e.g., other immune checkpoints) are not required to test the core model but may be explored as ancillary analyses without introducing new anchor genes.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CEACAM5, MSI-H
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CEACAM5: CEACAM5 is upregulated in bladder tumors and correlates with higher immune cell infiltration and MSI-H status, yet it is unknown whether this immune-rich state driven by MSI-H yields effective antitumor immunity or is functionally inhibited.

MSI-H: MSI-H and high immune infiltration are clinically most relevant to predicting response to immune checkpoint blockade; the central translational question is whether CEACAM5 modulates that response.

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions